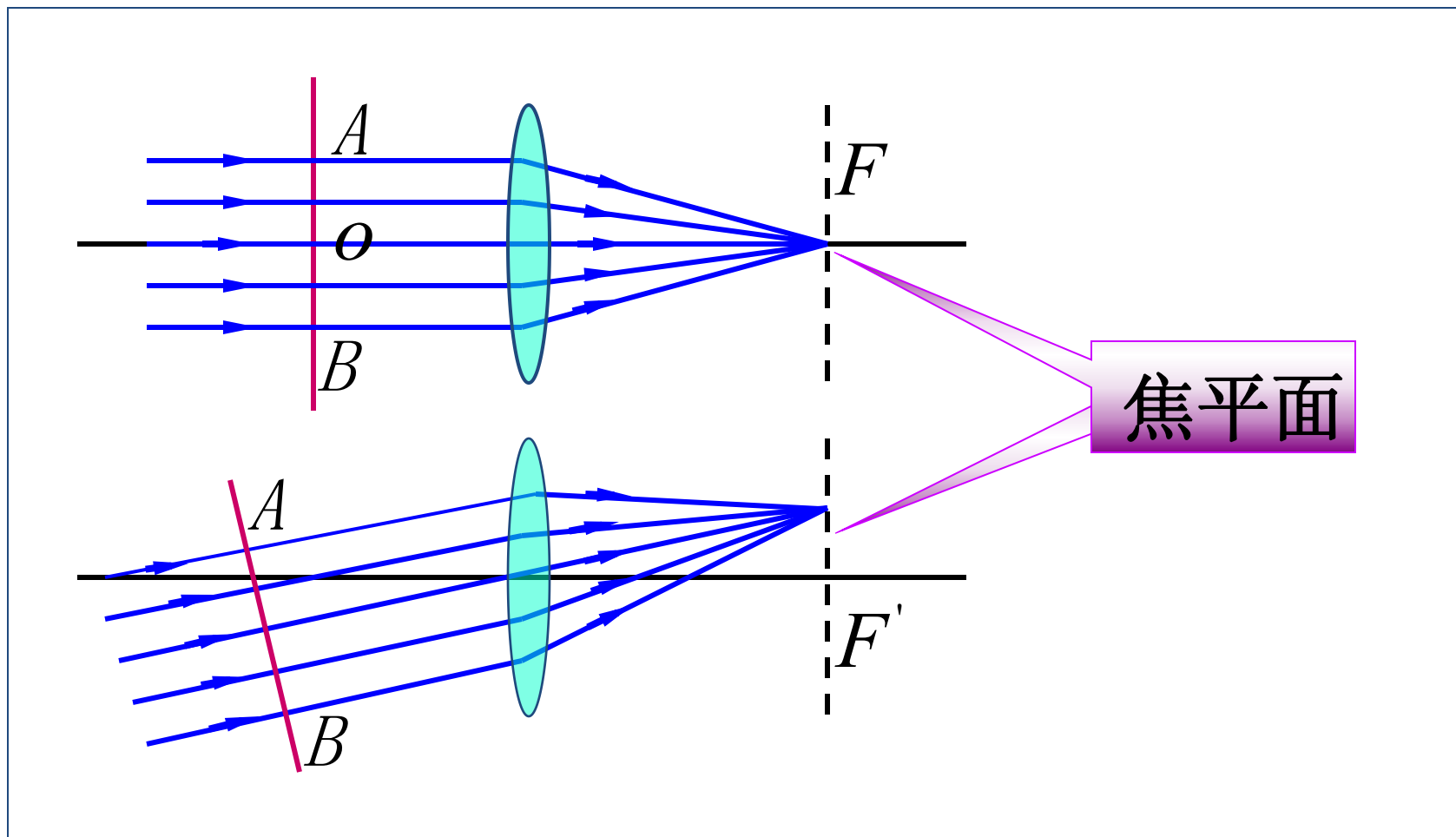




一 透镜不引起附加的光程差



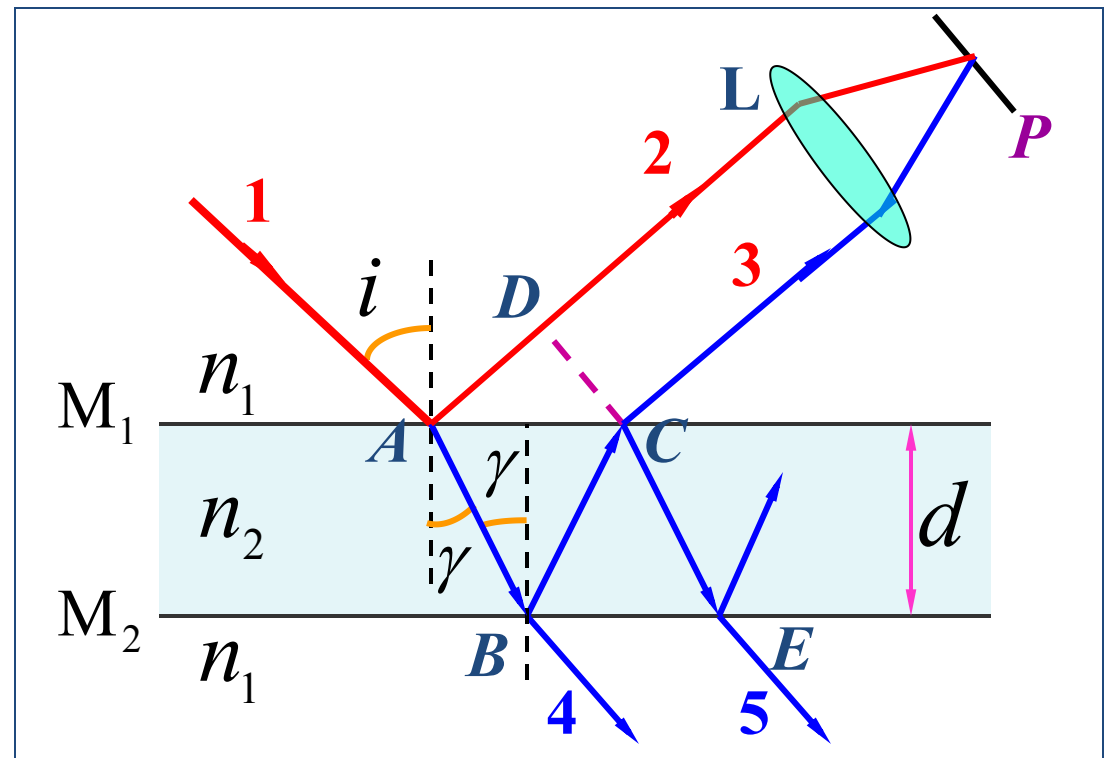


二 薄膜干涉的光程差

$$n_2 > n_1$$

$$CD \perp AD$$

$$\frac{\sin i}{\sin \gamma} = \frac{n_2}{n_1}$$





$$\Delta_{32} = n_2(AB + BC) - n_1 AD + \frac{\lambda}{2}$$

$$AB = BC = d / \cos \gamma$$

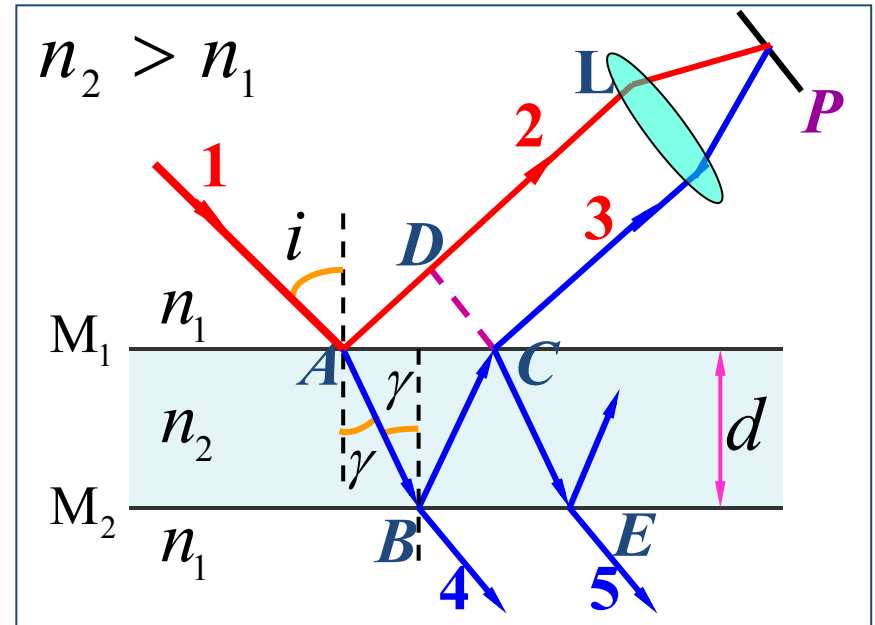
$$AD = AC \sin i = 2d \cdot \tan \gamma \cdot \sin i$$

$$\Delta_{32} = n_2(AB + BC) - n_1 AD + \frac{\lambda}{2}$$

$$= n_2 \left(\frac{2d}{\cos \gamma} \right) - n_1 \sin i 2d \tan \gamma + \frac{\lambda}{2}$$

$$\underline{n_1 \sin i = n_2 \sin \gamma \Rightarrow}$$

$$\Delta_{32} = \frac{2d}{\cos \gamma} n_2 - n_2 \sin \gamma 2d \frac{\sin \gamma}{\cos \gamma} + \frac{\lambda}{2} = \frac{2d}{\cos \gamma} n_2 (1 - \sin^2 \gamma) + \frac{\lambda}{2}$$

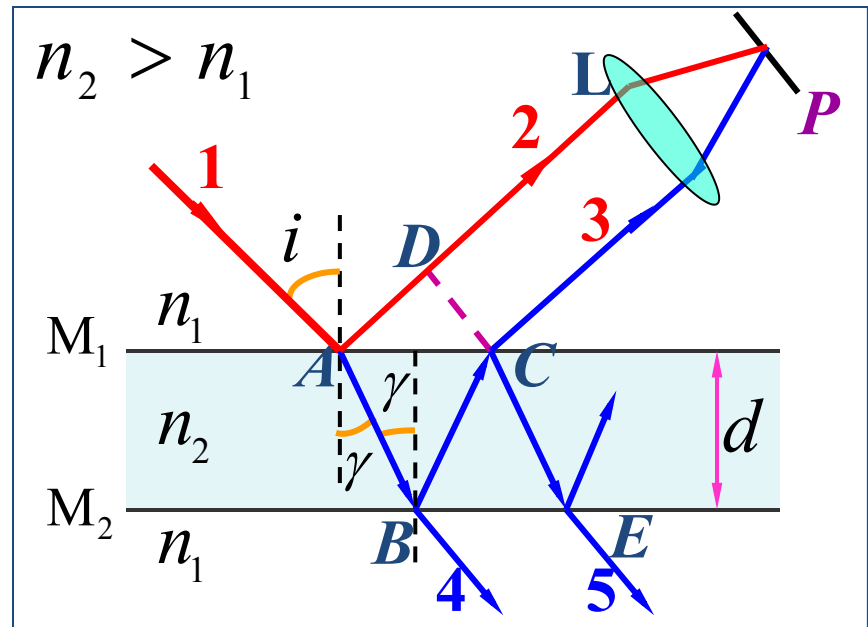




$$\Delta_{32} = \frac{2d}{\cos \gamma} n_2 (1 - \sin^2 \gamma) + \frac{\lambda}{2} = 2n_2 d \cos \gamma + \frac{\lambda}{2}$$

➤ 反射光的光程差 $\Delta_r = 2d \sqrt{n_2^2 - n_1^2 \sin^2 i} + \frac{\lambda}{2}$

$$\Delta_r = \begin{cases} k\lambda & \text{加强} \\ (k = 1, 2, \dots) \\ (2k+1)\frac{\lambda}{2} & \text{减弱} \\ (k = 0, 1, 2, \dots) \end{cases}$$





$$\Delta_r = 2d\sqrt{n_2^2 - n_1^2 \sin^2 i} + \lambda/2$$

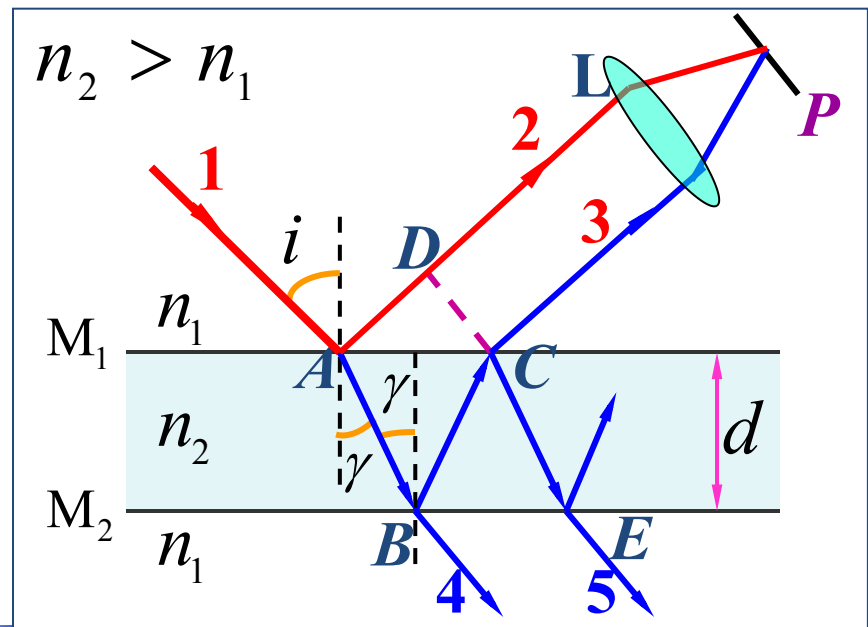
根据具体情况而定

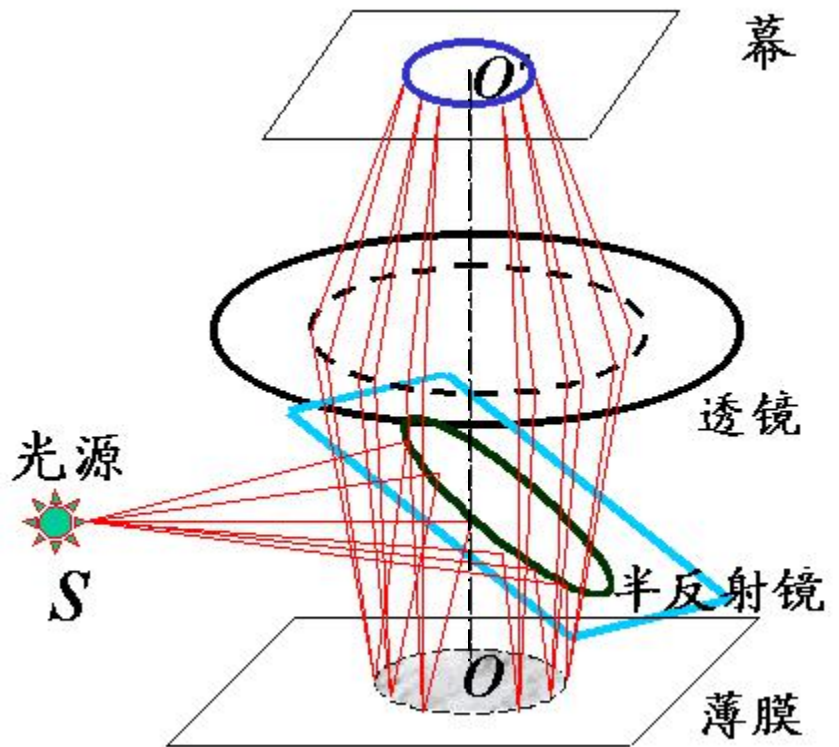
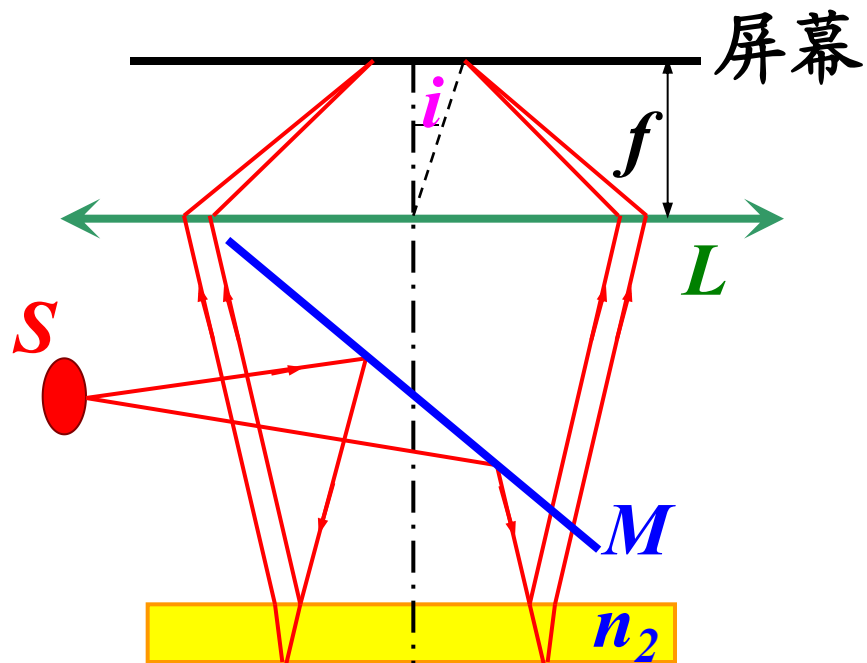
➤ 透射光的光程差 $\Delta_t = 2d\sqrt{n_2^2 - n_1^2 \sin^2 i}$

d 一定, i 同则 δ 同, 对应同一条条纹。

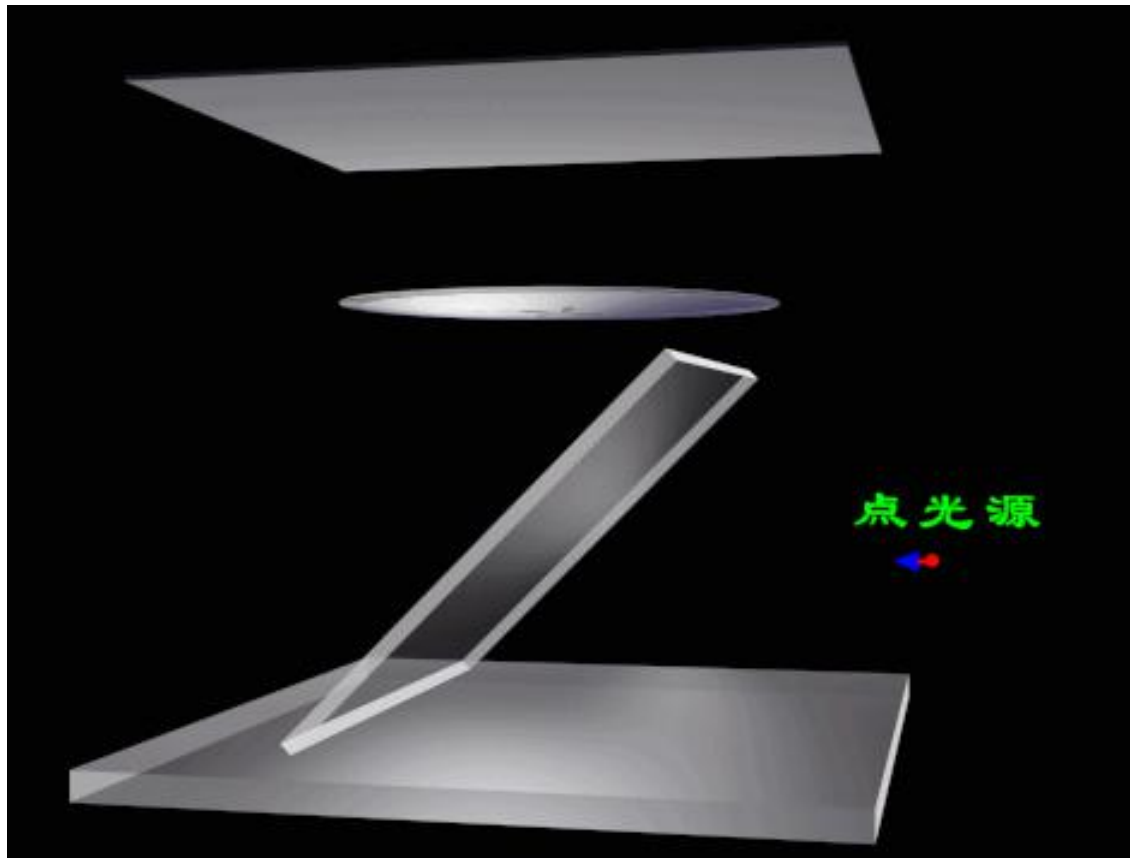
倾角相等的光线形成同一条条纹——等倾条纹。

注意：透射光和反射光干涉具有互补性，符合能量守恒定律。

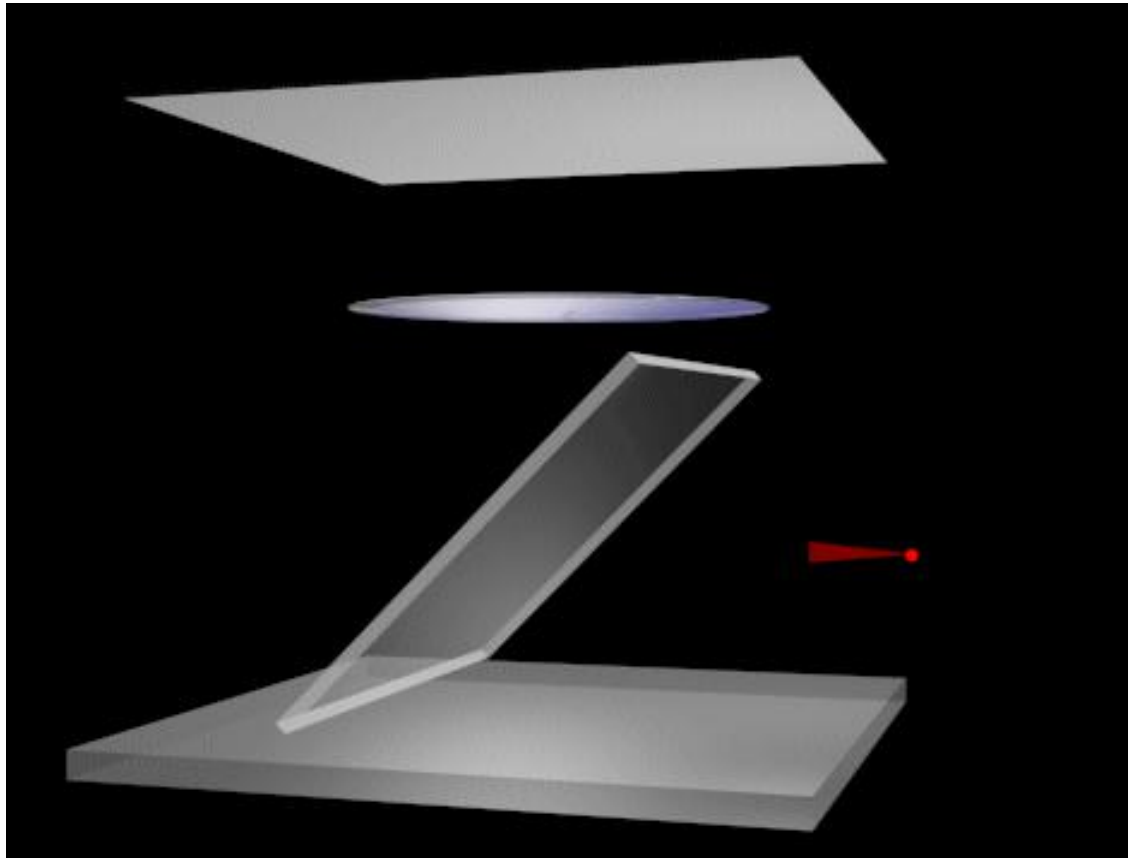




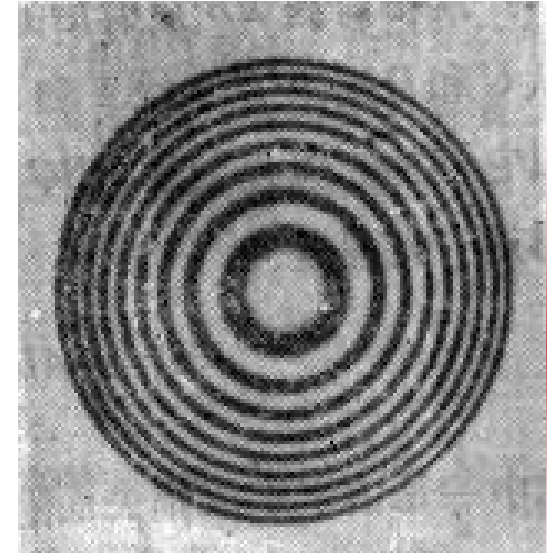
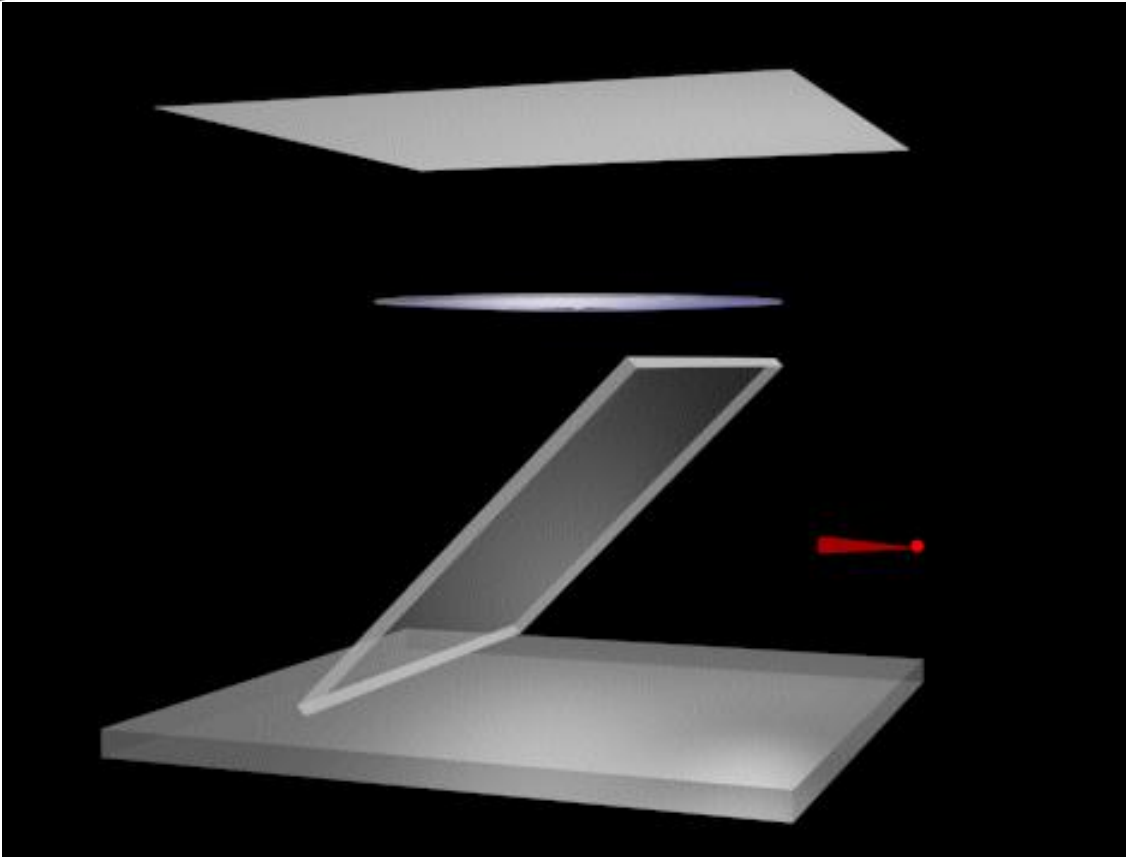
观察等倾条纹的实验装置和光路



从点光源发出的单条光线的光路



从点光源发出的锥面上光线的光路



等倾条纹照相

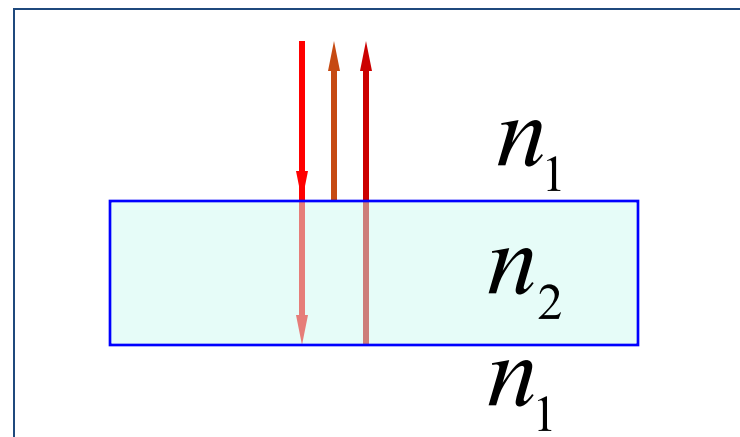
从点光源发出的锥体上光线的光路



◆ 当光线垂直入射时 $i = 0^\circ$

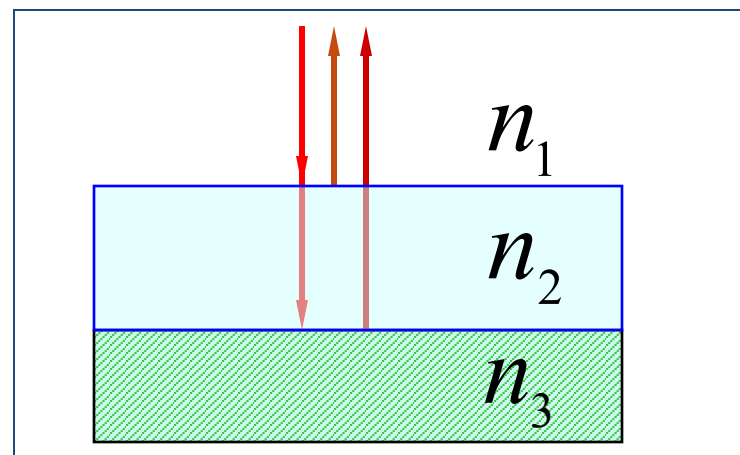
当 $n_2 > n_1$ 时

$$\Delta_r = 2dn_2 + \frac{\lambda}{2}$$



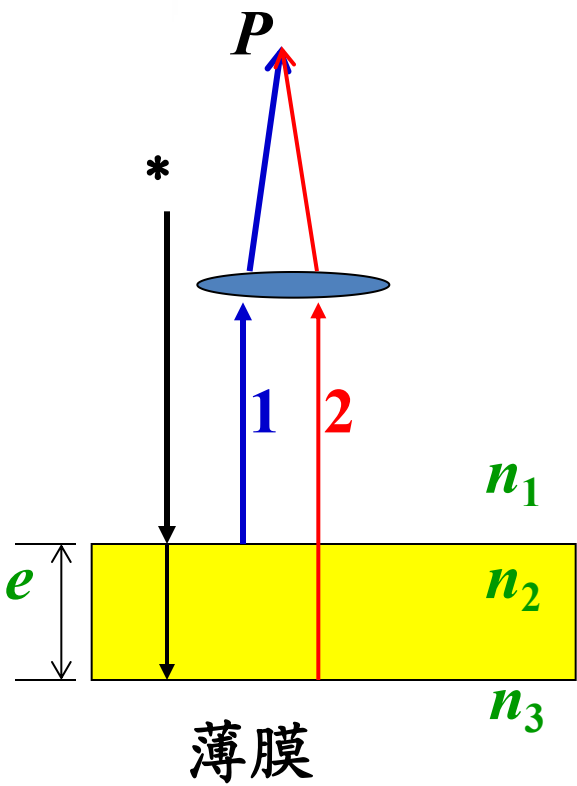
当 $n_3 > n_2 > n_1$ 时

$$\Delta_r = 2dn_2$$





讨论：薄膜干涉中反射光的半波损失



	反射光1 有无半 波损失	反射光2 有无半 波损失	反射光1、2 的光程差 δ
$n_1 < n_2 < n_3$	有	有	$2n_2e$
$n_1 > n_2 > n_3$	无	无	$2n_2e$
$n_1 < n_2 > n_3$	有	无	$2n_2e + \frac{\lambda}{2}$
$n_1 > n_2 < n_3$	无	有	$2n_2e + \frac{\lambda}{2}$

$n_1 < n_2 < n_3$ 或 $n_1 > n_2 > n_3$ ，不计

$n_1 < n_2 > n_3$ 或 $n_1 > n_2 < n_3$ ，计

斜入射
亦适用

或 $-\frac{\lambda}{2}$





例 一油轮漏出的油 (折射率 $n_1=1.20$) 污染了某海域, 在海水 ($n_2=1.30$) 表面形成一层薄薄的油污.

(1) 如果太阳正位于海域上空, 一直升飞机的驾驶员从机上向正下方观察, 他所正对的油层厚度为460 nm, 则他将观察到油层呈什么颜色?

(2) 如果一潜水员潜入该区域水下, 并向正上方观察, 又将看到油层呈什么颜色?



已知 $n_1=1.20$ $n_2=1.30$ $d=460 \text{ nm}$

解 (1) $\Delta_r = 2dn_1 = k\lambda$

$$\lambda = \frac{2n_1d}{k}, \quad k = 1, 2, \dots$$

$$k = 1, \quad \lambda = 2n_1d = 1104 \text{ nm}$$

$$k = 2, \quad \lambda = n_1d = 552 \text{ nm}$$

绿色

$$k = 3, \quad \lambda = \frac{2}{3}n_1d = 368 \text{ nm}$$



(2) 透射光的光程差 $\Delta_t = 2dn_1 + \lambda/2$

$$k = 1, \quad \lambda = \frac{2n_1d}{1 - 1/2} = 2208 \text{ nm}$$

紫红色 { $k = 2, \quad \lambda = \frac{2n_1d}{2 - 1/2} = 736 \text{ nm}$ 红光

$k = 3, \quad \lambda = \frac{2n_1d}{3 - 1/2} = 441.6 \text{ nm}$ 紫光

$$k = 4, \quad \lambda = \frac{2n_1d}{4 - 1/2} = 315.4 \text{ nm}$$

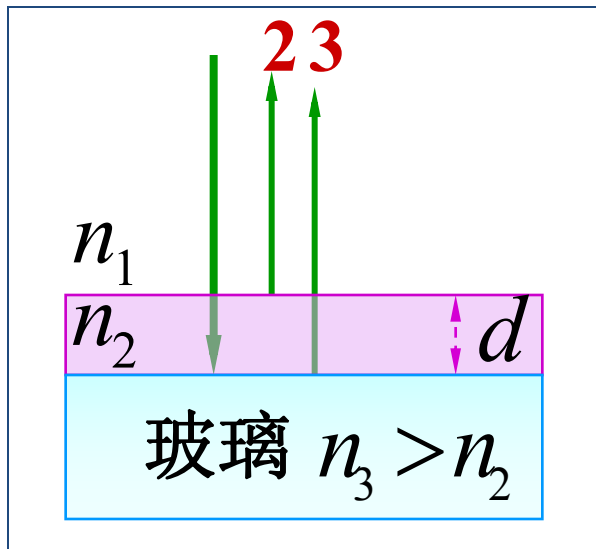


◆ 增透膜和增反膜

利用薄膜干涉可以提高光学器件的透光率。



例 为了增加透射率，求氟化镁膜的最小厚度．已知 空气 $n_1=1.00$ ，氟化镁 $n_2=1.38$ ， $\lambda=550\text{ nm}$



氟化镁为增透膜

解 $\Delta_r = 2dn_2 = (2k+1)\frac{\lambda}{2}$
取 $k=0$ **减弱**

$$d = d_{\min} = \frac{\lambda}{4n_2} = 99.6\text{ nm}$$

则 $\Delta_t = 2n_2d + \frac{\lambda}{2} = \lambda$ (**增强**)